



Department of Computer Science and Engineering  
Premier University

## CSE 460: Pattern Recognition Laboratory

Title: Bengali Digit Recognition: Implementation and Evaluation of a  
Convolutional Neural Network.

**Submitted by:**

|                        |                               |
|------------------------|-------------------------------|
| <b>Name</b>            | Mohammad Hafizur Rahman Sakib |
| <b>ID</b>              | 0222210005101118              |
| <b>Section</b>         | C                             |
| <b>Session</b>         | Fall 2025                     |
| <b>Semester</b>        | 8th Semester                  |
| <b>Submission Date</b> | 07.04.2026                    |

**Submitted to:**

Avishek Chowdhury  
Lecturer, Department of CSE  
Premier University, Chattogram

**Remarks**

# 1 Introduction

Handwritten digit recognition is a fundamental problem in the field of pattern recognition and computer vision. It has practical applications in areas such as document digitization, automatic number recognition, and financial systems.

In this project, a deep learning-based approach is employed to classify Bengali handwritten digits (0–9). The goal is to develop a robust model capable of accurately recognizing digit patterns from grayscale images. The model performance is evaluated using the macro F1-score, which ensures balanced evaluation across all classes.

## 2 Dataset Description

### 2.1 Dataset Summary

| Attribute         | Value              |
|-------------------|--------------------|
| Total Samples     | 36,682             |
| Number of Classes | 10 (0–9)           |
| Image Type        | Grayscale PNG      |
| Data Source       | CSV + Image Folder |

The dataset consists of grayscale images representing Bengali handwritten digits. Each image is associated with a label indicating the correct digit. The dataset is organized using CSV files that map image filenames to their corresponding labels.

## 3 Methodology

### 3.1 Data Preprocessing

To ensure consistent input and improve model performance, several preprocessing techniques were applied:

- Histogram equalization to enhance image contrast and highlight digit features
- Resizing all images to  $64 \times 64$  pixels for uniformity
- Normalization of pixel values to stabilize training
- Data augmentation techniques such as rotation and translation to increase dataset diversity

These preprocessing steps reduce noise, improve feature visibility, and help the model generalize better to unseen data.

### 3.2 Model Architecture

| Layer           | Configuration             | Purpose                       |
|-----------------|---------------------------|-------------------------------|
| Conv1           | 32 filters, $3 \times 3$  | Extract low-level features    |
| Conv2           | 64 filters, $3 \times 3$  | Capture intermediate patterns |
| Conv3           | 128 filters, $3 \times 3$ | Learn complex features        |
| MaxPooling      | $2 \times 2$              | Reduce spatial dimensions     |
| Fully Connected | 256 neurons               | Combine learned features      |
| Dropout         | 0.3                       | Prevent overfitting           |
| Output Layer    | 10 neurons                | Perform classification        |

The custom Convolutional Neural Network (CNN) extracts hierarchical features from the input images. Convolutional layers detect edges and patterns, while fully connected layers map these features to output classes.

### 3.3 Training Configuration

| Parameter     | Value              |
|---------------|--------------------|
| Optimizer     | Adam               |
| Loss Function | CrossEntropy Loss  |
| Learning Rate | $1 \times 10^{-3}$ |
| Batch Size    | 32                 |
| Epochs        | 15                 |

The Adam optimizer was chosen for its efficiency in handling large datasets. CrossEntropy loss is suitable for multi-class classification tasks.

## 4 Evaluation Metrics

The macro F1-score is used as the primary evaluation metric:

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

The F1-score balances precision and recall and is particularly useful when evaluating classification performance across multiple classes.

## 5 Results

### 5.1 Training Performance

| Epoch | Loss   | F1-score      |
|-------|--------|---------------|
| 1     | 1.0878 | 0.8969        |
| 2     | 0.4577 | 0.9370        |
| 3     | 0.3355 | 0.9482        |
| 4     | 0.2771 | 0.9533        |
| 5     | 0.2380 | 0.9665        |
| 6     | 0.2200 | 0.9646        |
| 7     | 0.1981 | 0.9664        |
| 8     | 0.1823 | 0.9697        |
| 9     | 0.1637 | 0.9741        |
| 10    | 0.1576 | 0.9733        |
| 11    | 0.1530 | 0.9741        |
| 12    | 0.1485 | 0.9708        |
| 13    | 0.1417 | <b>0.9774</b> |
| 14    | 0.1366 | 0.9742        |
| 15    | 0.1314 | 0.9735        |

The results show a steady decrease in training loss and a consistent improvement in validation F1-score, demonstrating effective learning and good convergence of the model. The best validation macro F1-score achieved is 0.9774 at epoch 13.

### 5.2 Classification Report

The classification report shows high precision, recall, and F1-score values for all digit classes. This indicates that the model performs consistently well across different classes without significant bias.

| Digit          | Precision | Recall | F1-score | Support |
|----------------|-----------|--------|----------|---------|
| 0              | 0.98      | 0.98   | 0.98     | 368     |
| 1              | 0.94      | 0.96   | 0.95     | 365     |
| 2              | 0.99      | 0.99   | 0.99     | 367     |
| 3              | 0.98      | 0.98   | 0.98     | 368     |
| 4              | 0.98      | 0.97   | 0.98     | 369     |
| 5              | 0.97      | 0.98   | 0.97     | 368     |
| 6              | 0.98      | 0.97   | 0.98     | 369     |
| 7              | 0.97      | 0.99   | 0.98     | 366     |
| 8              | 1.00      | 0.99   | 1.00     | 364     |
| 9              | 0.97      | 0.95   | 0.96     | 365     |
| <b>Overall</b> |           |        |          |         |
| Accuracy       | 0.98      |        |          |         |
| Macro Avg      | 0.98      | 0.98   | 0.98     | 3669    |
| Weighted Avg   | 0.98      | 0.98   | 0.98     | 3669    |

### 5.3 Confusion Matrix

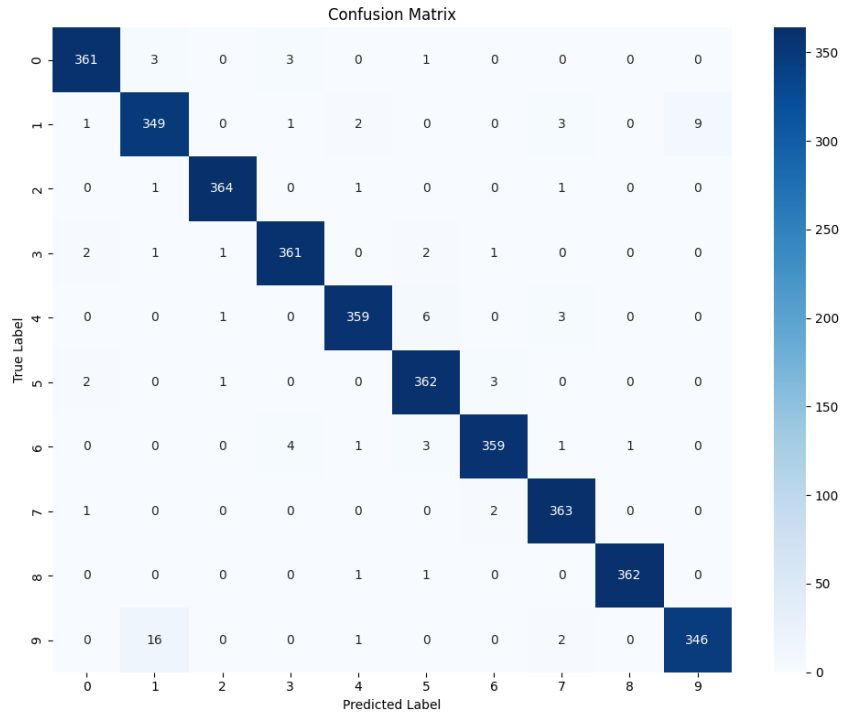


Figure 1: Confusion Matrix of the Proposed Model

The confusion matrix shows that most predictions are concentrated along the diagonal, indicating correct classification. Only minor misclassifications are observed between visually similar digits.

## 6 Conclusion

The custom CNN model developed in this competition achieved a strong macro F1-score of **0.9774** on the validation set, demonstrating excellent classification performance for Bengali handwritten digits. The combination of histogram equalization for contrast enhancement, data augmentation (rotation and translation), and the three-layer convolutional architecture played a vital role in improving feature extraction and model generalization. The network successfully captured both low-level edges and higher-level digit patterns, resulting in consistently high precision, recall, and F1-scores across all classes.

Overall, the proposed model proved highly effective for the task of Bengali handwritten digit recognition. Future work may focus on exploring deeper architectures such as ResNet, applying transfer learning from pre-trained models, implementing ensemble techniques, or incorporating more advanced augmentation strategies to further enhance performance and robustness.